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# Python Assignment 3: While Loop, For Loop
and Function
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import random

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# PART 1: Number Guessing Game (while loop,
else, break, continue)
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print("=" * 50)
print("NUMBER GUESSING GAME")
print("=" * 50)

# Generate a random number between 1 and 10
secret_number = random.randint(1, 10)
attempts = 3 # maximum number of valid
guesses allowed

while attempts > 0:
    try:
        guess = int(input("Guess the number

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(between 1 and 10): ")
    except ValueError:
        print("Please enter a whole
number.")
        continue

    # Out of range: inform the user and
    skip the rest of the loop
    # (this does not use up an attempt)
    if guess < 1 or guess > 10:
        print("Your guess is out of range.
Please guess a number between 1 and 10.")
        continue

    attempts -= 1

    if guess == secret_number:
        print("Congratulations! You guessed
the correct number.")
        break # exit
the loop on a correct guess
    elif guess > secret_number:
        print("Too high. Try again.")
    else:
        print("Too low. Try again.")
else:
    # Runs only if the loop ended without
    break (attempts used up)
    print("Better luck next time!")
    print("The correct number was:",
secret_number)

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# PART 2: Multiplication Table Generator
(for loop)
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print("\n" + "=" * 50)
print("MULTIPLICATION TABLE")
print("=" * 50)

number = int(input("Enter the number for
which you want the multiplication table:
"))

for i in range(1, 11):
    print(f"{number} x {i} = {number * i}")

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# PART 3: BMI Calculator (function)
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print("\n" + "=" * 50)
print("BMI CALCULATOR")
print("=" * 50)

def calculate_bmi(weight, height):
    """Return BMI = weight (kg) / height
(m) squared."""
    return weight / (height ** 2)

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weight = float(input("Enter your weight in  
kg: "))  
height = float(input("Enter your height in  
meters: "))  
  
bmi = calculate_bmi(weight, height)  
print(f"Your BMI is: {bmi:.2f}")
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